

Sales Analysis Project

Executive Summary

This project focuses on analyzing sales data to identify business trends, product performance, customer behavior, and regional profitability. The analysis was conducted using Python for data cleaning and exploratory analysis, Excel for dashboard creation, and Power BI for interactive business intelligence reporting.

The objective was to transform raw sales data into actionable insights that support data-driven decision-making.

Project Objectives

The main objectives of this project were:

1. Analyze overall sales performance.
2. Evaluate product profitability and demand.
3. Identify top-performing customers.
4. Compare sales and profit across regions.
5. Develop interactive dashboards for business reporting.
6. Generate insights that can improve business strategy.

Dataset Description

- Order ID
- Order Date

- Customer Name
- Product Name
- Category
- Sales
- Profit
- Quantity
- Region

These variables enable detailed analysis of revenue generation, profitability, customer purchasing patterns, and product demand.

Tools and Technologies

Python

- Data cleaning
- Missing value Analysis
- EDA
- Statistical summaries

Libraries used:

- Pandas
- NumPy
- Matplotlib

Microsoft Excel

Excel was used to create:

- KPI Dashboard
- Pivot Table
- Pivot Chart
- Interactive Slicer

Power BI

Power BI was used to develop:

- Interactive Dashboards
- KPI Cards
- Product Analysis
- Customer Analysis
- Regional Analysis

Data cleaning and Preparation

The following preprocessing steps were performed:

- Imported dataset into Python.
- Checked data types and column consistency.
- Removed duplicate records.
- Identified and handled missing values.
- Converted date fields into proper datetime format.
- Created calculated fields for reporting purposes.
- The cleaned dataset was then used for dashboard development.

```

print("Dataset Shape:", df.shape)

print("\nMissing Values:")
print(df.isnull().sum())

print("\nDuplicate Rows:", df.duplicated().sum())

df.drop_duplicates(inplace=True)
df.to_csv("amazon_cleaned.csv", index=False)

```

Dataset Shape: (51290, 10)

Missing Values:

Order ID	0
Customer ID	0
Product Name	0
Category	0
Quantity	0
Sales	0
Profit	0
Order Date	0
Region	0
State	0
dtype: int64	

Duplicate Rows: 9

Exploratory Data Analysis

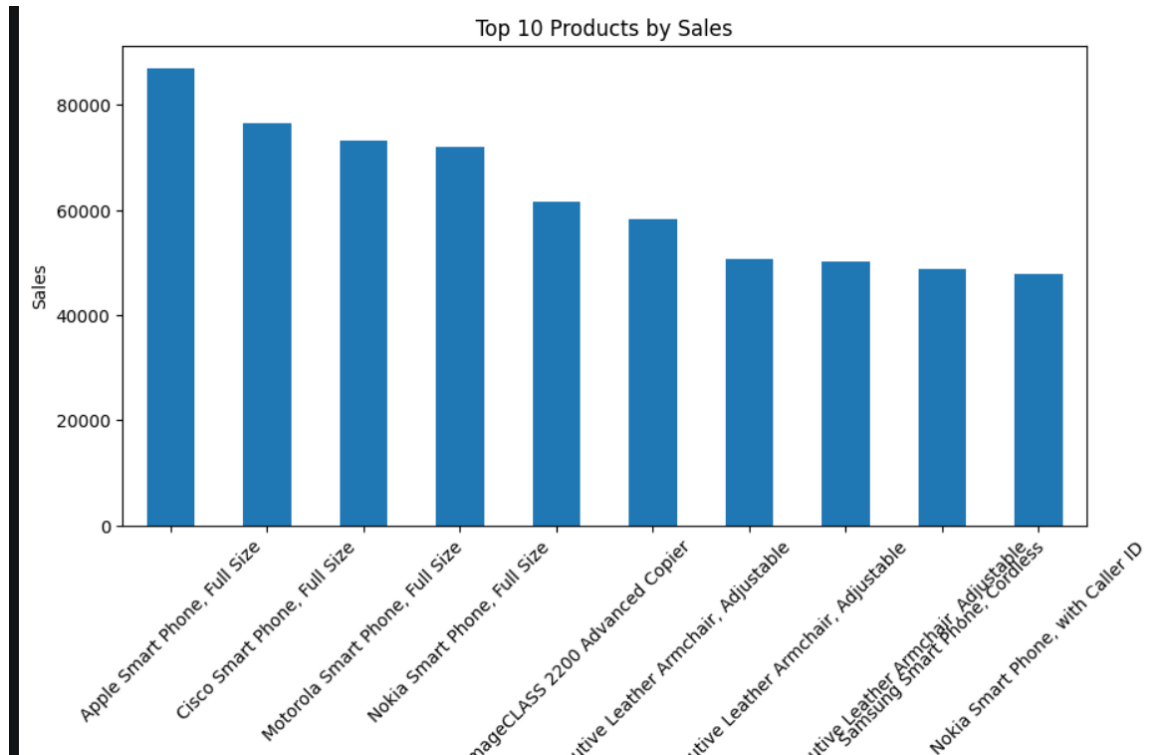
Several analyses were performed to understand business performance:

Sales trend were analyzed to determine:

- Revenue growth patterns
- Seasonal fluctuations
- High-performing periods

Product Analysis

- Top 10 product by sales
- Sales by category
- Quantity sold by product

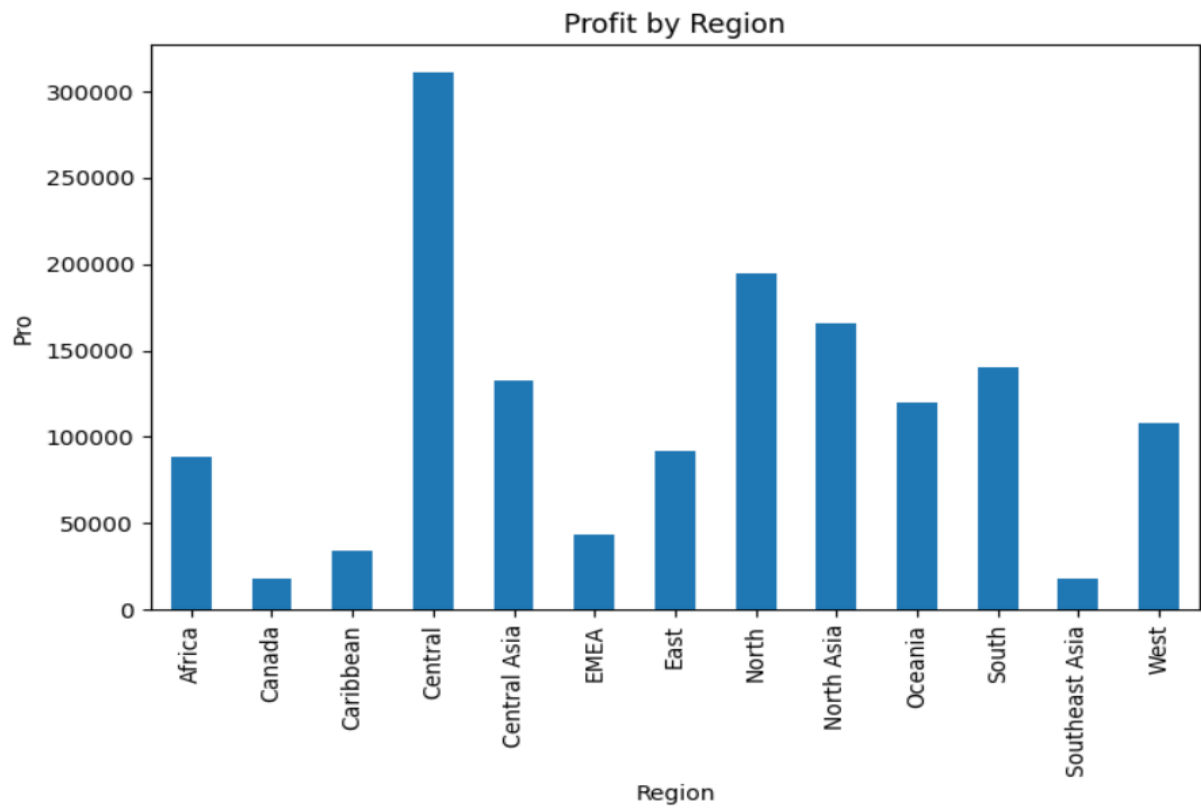
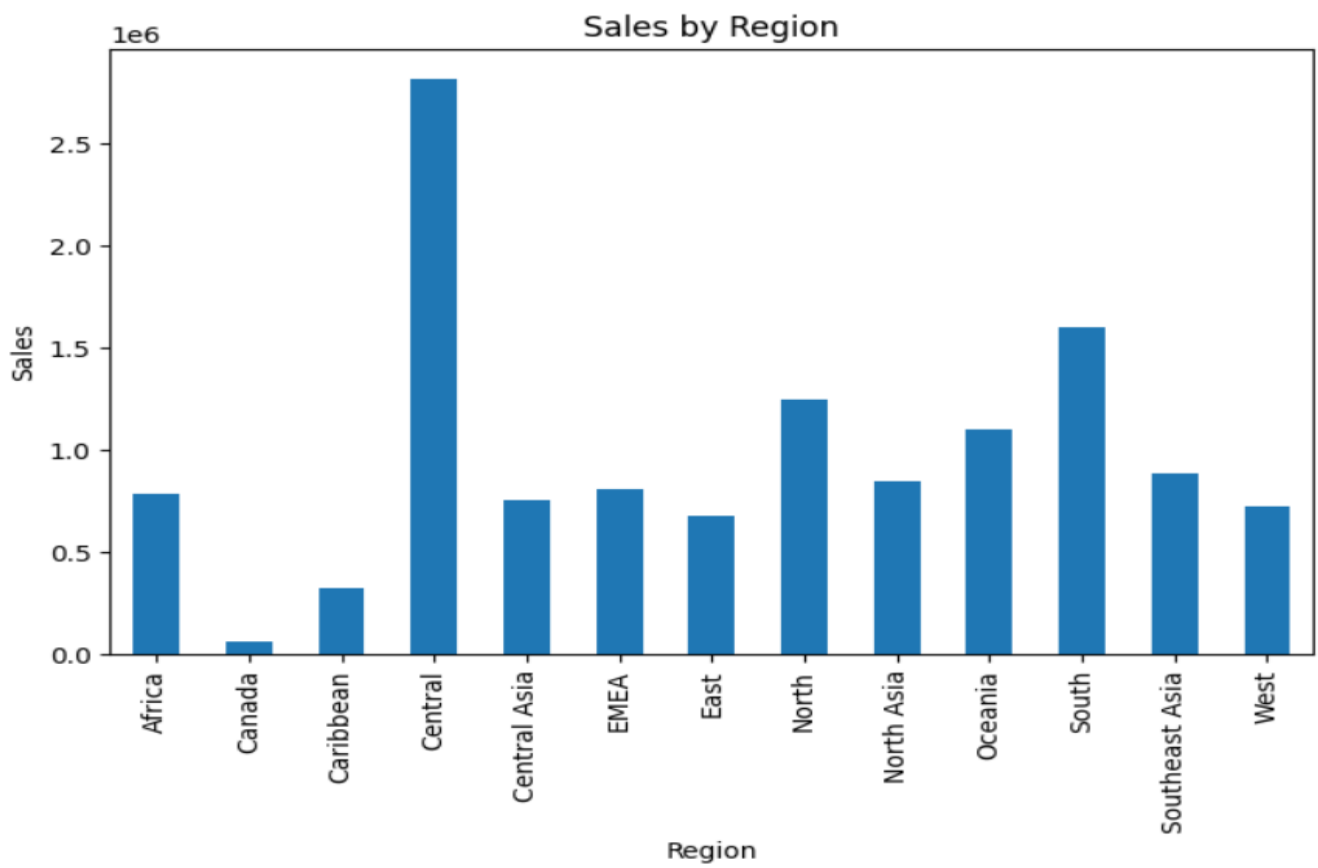


Regional Analysis

Regional performance was examined to

Determine:

- Sales by Region
- Profit by Region
- High-growth regions



Excel Dashboard

The Excel dashboard was developed using Pivot Tables and Pivot Charts.

Key Performance Indicators:

- Total Sales
- Total Profit
- Total Orders

Dashboard Components

- Sales by Category
- Sales by Region
- Profit by region

This dashboard enables quick monitoring of business performance.



Power BI Dashboard

Overview Dashboard

KIPs:

- Total Sales
- Total Profit
- Tital Orders
- Total Quantity

Visualizations:

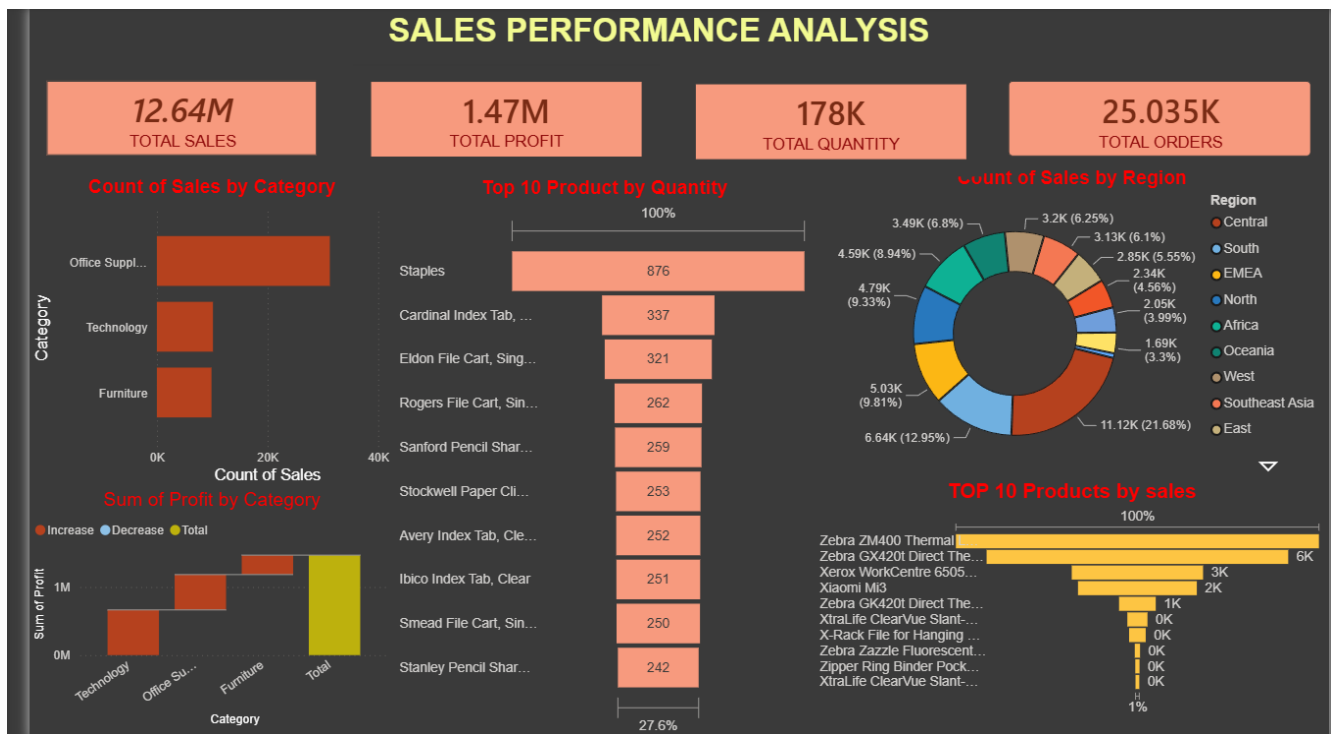
- Monthly Sales Trend
- Category Performance
- Regional Sales Distribution

Product Analysis Dashboard

- Top 10 Products by Sales
- Top 10 Products by Profit
- Quantity Sold by Product

Purpose:

- Identify best-selling products
- Evaluate profitability



Regional Analysis Dashboard

Visualization

- Top Customers by Sales
- Top Customers by Profit

Key Business Insights

The analysis revealed several important findings:

- A small number of products contribute a significant portion of total sales.
- High sales volume does not always translate into high profit.
- Certain customers contribute disproportionately to revenue generation.
- Regional performance varies significantly across markets.
- Product profitability should be monitored alongside sales performance

Business Recommendations

Based on the analysis, the following recommendations are proposed:

- Increase focus on high-profit products.
- Review pricing strategies for low-margin products.
- Strengthen relationships with top customers.
- Allocate resources toward high-performing regions.
- Use dashboard monitoring for continuous performance tracking.

Conclusion

This project demonstrates an end-to-end sales analytics workflow using Python, Excel, and Power BI. The combination of data cleaning, exploratory analysis, KPI reporting, and interactive dashboards provides valuable business insights that support strategic decision-making.

The project showcases essential Data Analyst skills including data preparation, visualization, business intelligence reporting, and insight generation.